Multiclass Classification of Gangrene and Ulcer Using MaxViT with Grad-CAM++ Explainability

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CERTIFICATE

This is to certify that this project report titled **"Multiclass Classification of Gangrene and Ulcer Using MaxViT with Grad-CAM++ Explainability"** is a bonafide record of work done by **P. SRI TEJA (228W1A12I2) and K. DEEKSHITHA (228W1A12F8)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V R Siddhartha Engineering College** (Autonomous under JNTUK) during the year **2025-26.**

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PROJECT SUMMARY

S. No	Item	Description
1	**Project Title**	**Multiclass Classification of Gangrene and Ulcer Using MaxViT with Grad-CAM++ Explainability**
2	**Student Names & Numbers**	Potluri Sri Teja (228W1A12I2) Kamisetty Deekshitha (228W1A12F8)
3	**Name of The Guide**	Dr. Y. Sangeetha
4	**Research Group**	Deep learning, Explainable AI.
5	**Application Area**	Health Care
6	**Aim of the Project**	To analyze medical images for classification into gangrene, ulcer and not ulcer or gangrene classes.
7	**Project Outcomes**	• Accurate classification of gangrene and ulcer skin disease. • Improved performance using MaxViT architecture. • Visual explanation of predictions using Grad-CAM++. • Support for clinical decision-making. • Robust performance on diverse medical images.

Student Signatures

1.

2.

Signature of the Guide

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TABLE OF CONTENTS

ACKNOWLEDGEMENT iii

LIST OF FIGURES v

LIST OF TABLES vi

ABSTRACT vii

CHAPTER-1 Introduction 01

1.1 Origin of the Problem 01

1.2 Basic definitions and Background 02

1.3 Problem Statement with Objectives and Outcomes 07

1.4 Applications 10

CHAPTER-2 Review of Literature 11

2.1Description of Existing Systems 11

2.2Summary of Literature Study 21

2.3 Software Requirement Specification 22

CHAPTER-3 Proposed Method 26

3.1Design Methodology 26

3.2System Architecture Diagram 29

3.3Description of Algorithms 34

3.4 Description of datasets and Tools 42

CHAPTER-4 Results and Observations 46

4.1Stepwise description of Results 46

4.2Result analysis and Test case results 49

4.3Observations from the work 58

CHAPTER-5Conclusion and Future study 59

5.1 Conclusion 59

5.2 Future study 60

REFERENCES

APPENDIX

LIST OF FIGURES

Figure	Page
Figure 3.1: Proposed MaxViT Architecture	30
Figure 4.1: Accuracy of MaxViT	49
Figure 4.2: Loss of MaxViT	50
Figure 4.3: Classification Report of the MaxViT model	50
Figure 4.4: Confusion Matrix for MaxViT Transformer	51
Figure 4.5: ROC curve of the MaxViT	51
Figure 4.6: Classification of Gangrene using MaxViT	52
Figure 4.7: Classification of Ulcer using MaxViT	53
Figure 4.8: Classification of Not ulcer or Gangrene using MaxViT	53
Figure 4.9: Original image and corresponding Grad-CAM++ heatmap for gangrene	54
Figure 4.10: Original image and corresponding Grad-CAM++ heatmap for ulcer	54
Figure 4.11: Original image and corresponding Grad-CAM++ heatmap for normal	55
Figure 4.12: User prediction with MaxViT model (gangrene)	55
Figure 4.13: User prediction with MaxViT model (ulcer)	56
Figure 4.14: User prediction with Grad-CAM++ XAI(not ulcer or gangrene)	56

LIST OF TABLES

Table Name	Page
Table 2.1: Summary of Prior Work	13
Table 4.1: Model Comparision of Medical Image Classification	57

ABSTRACT

Gangrene and ulcer are hard to tell apart at early stages, and misclassification delays the intervention window. This paper describes a MaxViT-based classifier that assigns medical images to one of three categories: gangrene, ulcer, or normal skin. MaxViT architecture can be used to capture local texture and global spatial structure in a single forward pass. This combination is useful here because the visual boundary between incipient gangrene and ulceration can be thin, especially for a model based predominantly on local features. A 70/15/15 split was adopted, with class balancing for uneven category distribution, and ImageNet pretraining with aggressive augmentation. Hold-out test set accuracy was 97.19%. Aggregate accuracy does not fully show failures by class, especially for a minority class. Therefore we also examined precision, recall and F1 per class, classwise ROC curve, and confusion matrix. We also used Grad CAM++ heatmaps to depict which parts of image the model paid more attention to during classification, enabling validation against spurious correlations. The model was then tested with images outside trained class to see if model could maintain performance in unseen environments.

Keywords: MaxViT, Medical Image Classification, Gangrene Detection, Ulcer Detection, Explainable AI (XAI), Grad-CAM++, Deep Learning, Transformer Models, Multiclass Classification, Healthcare AI.

CHAPTER-1
Introduction

1.1 Origin of the Problem

In cases of gangrene and ulcer, the seriousness of the condition is usually downplayed during its early stages of development. However, the necrosis of tissues and the infections that occur progress rapidly, and the treatments employed during the first few days after the disease onset may not be effective when used weeks after [1]. In practice, however, diagnosis is primarily made through visual inspection, which is reliable under ideal circumstances but highly susceptible to error on a mass scale due to patient fatigue and heavy workloads [2].

The development of deep learning algorithms has made significant strides in the area of automatic classification. CNNs were among the earliest models for identifying skin lesions and analyzing wounds, while transformers took the idea further by integrating the concept of attention, which allows models to recognize structures in long ranges that convolutional algorithms cannot see [3], [4]. The combination of convolutional and vision transformers models, or even MaxViT, has drawn special attention since it allows both local texture recognition and contextual relationships to be handled at once. This approach can be useful in situations where visual presentations vary considerably between cases [5], [6].

The interpretability problem has not kept up. Most high performing models produce a prediction with no account of what produced it. In many domains that is an acceptable trade-off. In clinical settings it is harder to defend—a prediction a clinician cannot interrogate is one they cannot safely use. Grad-CAM and its variants attempt to close that gap by generating spatial maps that show which image regions drove a given classification, giving practitioners something concrete to evaluate rather than a probability score alone [7], [8].

New research works tend to explore architectures that attempt to achieve both accuracy and explainability simultaneously. Combinations like CNN-ViT, ConvNeXt-MaxViT hybridization, and other parameter-efficient transformer architectures have shown that explainable spatial reasoning could be accomplished without sacrificing discriminative capacity significantly, indicating that the accuracy-explainability trade-off is not an intrinsic issue [9], [10]. Additional research supports the utility of deep learning algorithms for image classification tasks in medicine, specifically the use of CNNs for diabetic foot ulcer recognition [11]. The advancements made in the field of explainable artificial intelligence have highlighted the

significance of introducing interpretation tools like those that work on the basis of Grad-CAM [12].

This research proposes an explainable classification scheme based on the MaxViT neural network architecture to classify tri-class images as either gangrene, ulcer, or neither gangrene nor ulcer. This classification approach is designed using pre-trained weights from ImageNet using stratified data splitting and balanced classes due to an uneven distribution of categories. Grad-CAM++ tool is used to generate explanations and improve the interpretability of the approach.

1.2 Basic Definitions and Background

This subsection provides an overview of the theoretical basis of the classification methodology proposed in this study. It covers issues related to the disease states being studied, the use of deep learning methods in medical imaging, and specific architecture designs and interpretability techniques that the methodology relies on. Awareness of these terms is crucial for comprehending the methodology proposed in this study.

1.2.1 Basic Definitions

Gangrene

Gangrene kills tissue. That's the blunt version. Blood supply cuts off, bacteria move in, or both happen at once — and the tissue dies. It tends to show up in the extremities first: toes, feet, fingers. People with diabetes or peripheral artery disease are particularly vulnerable because their circulation was already struggling before any of this started.

Three types show up in clinical settings. Dry gangrene is the slow one — blood flow drops, no infection, tissue gradually dies back. Wet gangrene is the one you don't want: bacteria involved, spreads quickly, much harder to get ahead of. Gas gangrene is rare, but rare doesn't mean safe. Gas-producing bacteria can make it life-threatening in hours. The timeline is what makes it serious.

The tissue gives itself away visually. Dark brown to black discoloration, visible decay, skin that's broken down. You can see it — which is exactly why image-based detection keeps coming up as a practical option. The catch is that it only works if the model has been trained on how gangrene actually looks across different skin tones and at different stages of infection. A model trained on a narrow slice of presentations will miss cases it hasn't seen.

The consequences of getting it wrong aren't abstract. Catch it late and amputation is probably coming. Miss it entirely and systemic infection follows. There isn't much room between those two outcomes.

Ulcer

An ulcer is an open wound in the skin. Pressure, poor circulation, and diabetes are the most common causes — and in the case of diabetic foot ulcers, the consequences can be severe. They are a leading driver of both hospitalization and amputation, which is reason enough to take classification seriously. Visually, ulcers present as inflamed or reddish lesions, sometimes with discharge and tissue damage at varying depths. Unlike gangrene, they do not begin with tissue death — but untreated ulcers get there. That progression is also what makes them difficult to classify manually. At certain stages, an ulcer and early gangrene look similar enough that visual inspection under clinical time pressure becomes genuinely unreliable. Automated classification exists precisely to cover that gap. A model trained on sufficient variation in wound depth, skin tone, and infection stage can catch distinctions that a rushed clinical eye will sometimes miss.

Normal (Not Ulcer or Gangrene) Condition

The normal class contains images of healthy skin — no ulceration, no gangrene, no visible pathology. Its role in the dataset is often underappreciated. Without healthy baseline images, a classifier has no reference point for what non-diseased tissue actually looks like. It can only learn to distinguish one condition from another, not conditions from their absence.

Including normal samples directly reduces false positives. In a medical context, that is not a minor technical detail. Flagging healthy skin as diseased carries real consequences — unnecessary intervention, patient anxiety, and added pressure on clinical workflows that are already stretched. A model that cannot confidently identify normal tissue is not ready for deployment, regardless of how well it performs on the pathological classes.

Medical Image Classification

Medical image classification sorts images into diagnostic categories using computational models. In practice, it gives clinicians a second read on visual data that would otherwise rely entirely on human judgment. Earlier methods extracted features by hand - texture analysis, edge detection, handcrafted descriptors. They worked up to a point but fell apart when image quality varied or conditions presented unusually. Deep learning removed that ceiling by learning

features directly from data, which is why classification performance improved as sharply as it did across most imaging tasks.

Deep Learning for Image Analysis

Deep learning uses multi-layered neural networks to learn representations of data at increasing levels of abstraction - from edges and textures at the lower layers to semantic patterns higher up. That hierarchical structure is what makes it well-suited to image analysis.

CNNs are the standard architecture for image classification. They extract spatial features efficiently, but their receptive fields are local by design. What a CNN sees at any given point in an image is a small neighborhood - it does not naturally relate distant regions to one another. In medical imaging, that limitation matters: diagnostic signals are often distributed across an image rather than concentrated in one patch.

MaxViT Architecture

MaxViT (Multi-Axis Vision Transformer) combines convolutional and transformer-like attention modules in one architecture. It is quite intuitive because, on the one hand, CNNs effectively extract low-level spatial features, but do not take into account the interrelationships between distant objects, and, on the other hand, transformers work well for modeling global relations; however, scaling up their size becomes inefficient when working with higher-resolution images.

The first step in MaxViT involves convolutional layers, which extract fine-grained local features such as textures, shapes, and contours. Next, the multi-axis self-attention layer connects image regions that could not have been linked using local convolutions. Instead of performing global attention on all pixels simultaneously, which is computationally costly, the model applies attention operations across several axes sequentially, thus ensuring efficient computation while reasoning over the whole image.

This difference is crucial when differentiating between gangrene and ulcers. These pathological conditions share the same visual characteristics—discoloration, uneven skin surface, damaged tissues—and the distinction is only possible by analyzing the textures of the lesions and their relation to each other. The classifier based only on local information tends to fail in cases where some elements are missing, while the system that analyzes only global relations may classify gangrene as an ulcer and vice versa.

Transfer Learning

Transfer learning starts with a model already trained on a large dataset and adapts it to a new task rather than training from scratch. Here, the MaxViT model is initialized with weights pretrained on ImageNet. For medical imaging, this is a practical necessity more than an optimization. Labeled medical datasets are expensive to build and rarely large enough to train a deep architecture from random initialization without overfitting. Pretrained weights give the model a starting point - feature detectors that already respond to edges, textures, and shapes - which the fine-tuning process then adjusts toward the specific demands of gangrene and ulcer classification.

Data Augmentation

Data augmentation expands the training set through transformations - rotation, flipping, scaling, brightness adjustment - that reflect variation in real clinical photographs. No additional labeled data is needed; the existing set just covers more ground.

This matters in medical imaging because large annotated datasets are rare. A small, uniform training set produces a model that memorizes rather than generalizes. Augmentation breaks that pattern by exposing the model to enough variation that it cannot rely on surface-level cues alone.

Explainable AI using Grad-CAM++

Deep learning models used in clinical contexts require some degree of interpretability. A model that classifies an image correctly but offers no indication of what drove that classification is difficult to validate, and difficult to trust in practice. Grad-CAM++ was applied in this study to examine whether the network's predictions correspond to clinically meaningful image regions.

Grad-CAM++ computes gradients flowing into the final convolutional layers and uses them to assign spatial weights across the input image. The method extends standard Grad-CAM by handling multi-region cases more accurately—when more than one area of an image is diagnostically relevant, Grad-CAM++ localizes each of them more precisely than its predecessor.

The heatmaps are examined to assess whether the model attends to infected tissue and wound areas or to irrelevant features such as background variation or image artifacts. This does not resolve the interpretability problem in any complete sense, but it provides clinicians with a concrete means of checking model behavior on individual cases—something that aggregate

performance metrics cannot offer.

Evaluation Metrics

Model performance is evaluated using classification metrics. Accuracy reflects overall classification correctness, but per-class precision and recall are more informative when class sizes differ or when the cost of misclassifying one category—such as missing a gangrene diagnosis—is higher than another. F1-score consolidates both into a single per-class measure, making it useful for comparing performance across an imbalanced dataset.

Confusion matrices were examined to identify which class pairs the model confuses most frequently. ROC curves provided an assessment of discrimination ability across decision thresholds, independent of any single operating point. Using these tools together gives a clearer account of model behavior than aggregate accuracy alone.

1.2.2 Background

Interpretation of medical imaging is one of the crucial aspects of contemporary medicine since it helps doctors identify diseases faster and more accurately than they would using their naked eyes. Skin conditions like gangrene and ulcers are particularly time-sensitive. Catch them early and you have options. Miss them, and you're looking at spreading infection, serious tissue damage, and a much harder recovery. The traditional approach — a clinician eyeballing the wound — works, but it's imperfect. Experience levels vary. Lighting varies. Patients vary. What one doctor catches, can be missed by the other.

AI and machine learning found their way into medical imaging almost inevitably — the data was there, the problem was real, and the old methods had obvious ceilings. Early systems relied on handcrafted features: texture, color, edge detection. Researchers manually told the model what to look for. It worked, up to a point. But medical images are messy and varied, and these approaches broke down when conditions didn't fit the expected pattern.

CNNs changed that. Instead of being told what features matter, the model learned them directly from data. They got very good at picking up local patterns — edges, textures, fine details. The catch is they struggle to see the bigger picture. A CNN might nail a lesion's texture while missing how it relates to surrounding tissue. Transformers were built to fix exactly that. Using self-attention, they look across the entire image at once, capturing relationships between distant regions. Vision Transformers (ViTs) brought this to medical imaging — but they're hungry.

They need large datasets and serious compute, which hospitals don't always have.

Hybrid models like MaxViT split the difference. CNN layers handle the local detail work; transformer layers handle the broader context. For something like distinguishing gangrene from ulcers from normal skin, that combination matters — you need both. There's still the black-box problem, though. A model that's right 95% of the time but can't explain itself is a hard sell in a clinical setting. Grad-CAM++ helps by generating heatmaps — visual overlays showing which parts of the image actually drove the decision. It doesn't fully solve interpretability, but it gives clinicians something to look at and question, which is a start.

1.3 Problem Statement

Gangrene and ulcers don't wait. Both follow the same upstream causes — diabetes, poor circulation, prolonged pressure — and without early intervention, the path is predictable and grim. Missed gangrene means tissue dies. A missed ulcer goes deep. Either way, you're looking at amputation, or worse.

Right now, diagnosis comes down to a clinician examining the wound. When the right specialist is in the room, that works reasonably well. When they're fatigued, overloaded, or simply unavailable — which describes most of the world — it doesn't. Specialist access is scarce in many regions, and early detection pays the price.

The conditions themselves don't make it easier. Early-stage gangrene and ulcers look similar: discoloration, inflammation, surface irregularity. The differences are real, but narrow enough that experienced clinicians disagree. Add inconsistent image quality and variable lighting, and the margin for error widens further.

Deep learning has helped, but not enough. CNNs are good at reading local texture and poor at understanding how parts of the image relate to each other — a genuine problem when diagnosis depends on both at once.

Then there's interpretability, which is an older and harder issue. Models return prediction without explanation. In computer vision, that is workable. In clinical practice it is not. A clinician who cannot trace a prediction cannot validate it, and one they cannot validate does not change how they treat a patient. Accuracy alone does not drive adoption.

The framework proposed here uses MaxViT — combining local convolutional feature extraction with multi-axis self-attention — to handle both demands within a single model. Grad-CAM++

generates heatmaps identifying which image regions drove each decision. The goal is a system clinicians can rely on and interrogate.

1.3.1 Objectives

The aim of the current research is to create a precise, automatic and explainable medical image classification system for identifying severe skin diseases using modern deep learning approaches. The research aims to achieve the following objectives:

- Creating and implementing the Automated Medical Image Classification System: To design an advanced model that will allow processing medical images using steps such as preprocessing, feature extraction, classification, and evaluation to classify images that contain gangrene, ulcers, and healthy (non-ulcer and non-gangrene) cases.

- Creating an Effective Preprocessing Algorithm: To apply various preprocessing methods including data cleaning, data splitting, data normalization, and data resizing to achieve data uniformity and ensure its quality.

- Applying Data Augmentation Method to Enhance Generalization Ability: To use data augmentation techniques such as rotating, flipping, zooming, and brightness correction to increase diversity in datasets and ensure the reliability of the model.

- Applying Advanced Neural Network Architecture (MaxViT): To employ MaxViT (Multi-axis Vision Transformer) architecture that allows capturing both global and local features to ensure effective image classification.

- Applying Transfer Learning Approach: To utilize transfer learning to increase learning efficiency, reduce learning times and ensure more accurate results.

- Ensuring Balanced Learning by Applicating Class Weights: To use class weights for mitigating potential problems associated with class imbalance.

- Implementing Explainable AI Methods (Grad-CAM++): To utilize Grad-CAM++ method for visual explanation generation.

- Evaluating the Quality of the Model: To assess the model's quality using standard metrics such as accuracy, precision, recall, F1-Score, confusion matrix, and ROC Curve.

1.3.2 Expected Outcomes

The study anticipates the following outcomes.

- High-Accuracy Multiclass Classification: The system achieves 97.19% overall classification accuracy across the three categories - gangrene, ulcer, and normal skin. This level of performance indicates that the model can distinguish between conditions that present with considerable visual overlap, which is the central challenge of the classification task.

- Improved Feature Representation Using MaxViT: The MaxViT architecture extracts features at multiple scales, capturing both fine-grained texture and broader lesion structure. This dual sensitivity is what allows the model to separate visually similar classes more reliably than single-scale approaches.

- Consistent Performance Across Image Conditions: Preprocessing and augmentation reduce the model's sensitivity to lighting variation, background clutter, and texture differences. The result is classification performance that holds across images collected under different clinical conditions, not just those resembling the training set.

- Interpretable Predictions with Grad-CAM++: Grad-CAM++ produces heatmaps that show which image regions influenced each prediction. Clinicians can use these to check whether the model is attending to pathologically relevant areas—infected tissue, wound margins—rather than incidental features. This does not make the model fully transparent, but it gives practitioners a concrete way to interrogate individual outputs.

- Efficient Training: Transfer learning is combined with differentiated learning rates and GPU acceleration, reduces the number of epochs required to reach convergence. The training setup direct to reproduce, and the methodology transfers to other dermatological datasets without substantial modification.

- Foundation for Clinical and Applied Extensions: The framework is not limited to the three conditions examined here. The architecture can be extended to other dermatological conditions, and the system is compatible with integration into clinical decision support tools or web and mobile platforms. Each of these extensions would require independent validation before deployment, but the current design provides a concrete methodological basis for that work.

1.4 Applications

The framework has several practical applications in clinical and healthcare contexts.

- In diagnostic settings, automated image classification paired with Grad-CAM++ heatmaps gives clinicians a way to check what the model is responding to in each image. Rather than accepting or rejecting a prediction without basis, a clinician can see whether the highlighted regions correspond to actual pathological features—infected tissue, wound margins, necrotic areas—before acting on the output.

- The model can be used as a screening tool for diabetic patients, delay in detection of gangrene or ulcers can allow infection to spread rapidly and broadly. Catching suspicious and care needed cases earlier, even with imperfect accuracy, moves the window for intervention to a point where outcomes are more manageable.

- In telemedicine, patients with limited access to specialist care could submit images remotely for preliminary classification. Web and mobile deployment extends this further, allowing image capture and initial assessment before a hospital visit. These are not substitutes for clinical evaluation, but they can shorten the gap between symptom onset and professional assessment—which, in conditions like gangrene, is not a trivial consideration.

- At the hospital level, integration into existing image analysis pipelines allows the framework to process routine classification at scale, with outputs that are more consistent than manual review across large caseloads and shift changes. The Grad-CAM++ visualizations carry additional value in training: students and trainees can examine which regions the model attends to and compare that against their own developing diagnostic reasoning.

- Future work can extend the architecture to additional dermatological conditions and refine the interpretability component. Both are worth pursuing—classification scope and model transparency remain the two areas where the gap between research systems and clinical deployment is hardest to close.

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

The literature on satellite image analysis demonstrates significant progress in land- cover classification, deep-learning architectures, and change-detection frameworks. The following section reviews six key systems that form the foundation for this project.

[1] An Image-Based Gangrene Disease Classification

Authors: P. S. Nair, T. A. Berihu, V. Kumar
Published Year: 2020

Observations:

This study focuses on the early-stage classification of gangrene using image-based analysis, emphasizing the importance of timely detection to prevent severe medical complications. The authors employ traditional machine learning techniques, where features are manually extracted from images based on visible characteristics such as tissue discoloration, necrotic regions, texture variations, and patterns of infection spread. These handcrafted features are then used to train a classification model capable of distinguishing gangrene from non-gangrene cases.

A key contribution of this work is the demonstration that automated image-based diagnosis is feasible and can reduce reliance on manual clinical evaluation. The proposed system offers a more consistent and structured approach to detection, helping to minimize variability caused by human judgment. This is particularly useful in scenarios involving large-scale screening or limited access to experienced medical professionals, where automated assistance can improve efficiency and early diagnosis.

However, the study also highlights important limitations of traditional machine learning approaches. The model's performance is heavily dependent on the quality of handcrafted features, which may not capture the complex and diverse patterns present in real clinical images. Variations in lighting, image quality, and patient conditions further reduce robustness and generalization. The authors acknowledge these challenges and suggest that deep learning methods, which can automatically learn hierarchical features from data, are better suited for handling such complexity and improving overall classification accuracy.

[2] Advances in Image-Based Diagnosis of Diabetic Foot Ulcers Using Deep Learning Techniques

Authors: H. F. Alhasson, M. Alshammari, A. Alshammari
Published Year: 2025

Observations:

This paper provides a comprehensive review of CNN-based architectures for the diagnosis of diabetic foot ulcers, with a particular focus on their ability to identify ulcer severity and monitor disease progression over time. The study highlights how deep learning models automatically extract hierarchical features from medical images, enabling more accurate detection compared to traditional machine learning approaches. CNNs are shown to be effective in capturing local visual patterns such as texture variations, wound boundaries, and color differences, which are critical indicators in ulcer analysis.

The review clearly establishes that deep learning methods significantly outperform conventional techniques in terms of accuracy and reliability. However, it also emphasizes persistent challenges in this domain. One major issue is dataset imbalance, where certain classes (e.g., severe ulcers) are underrepresented, leading to biased model predictions. Additionally, the visual appearance of wounds varies widely due to differences in lighting conditions, infection stages, and patient-specific factors, making consistent classification difficult. These variations reduce the robustness and generalization capability of CNN-based systems when applied to real-world clinical data.

Another important limitation discussed in the paper is the lack of interpretability in CNN models. While they achieve strong performance, they function as black-box systems, providing predictions without explaining the reasoning behind them. This lack of transparency is a significant concern in medical applications, where clinicians require clear justification for automated decisions. The study concludes that although CNNs provide a solid foundation for medical image classification, they struggle to capture global contextual relationships and fail to offer explainable outputs, highlighting the need for more advanced architectures and explainable AI techniques.

[3] Enhanced Tuberculosis Detection Using Vision Transformers and Explainable AI

Authors: K. Vanitha et al.
Published Year: 2025

Observations:

This study explores the application of Vision Transformer (ViT) models for tuberculosis detection using chest X-ray images, combined with explainable AI techniques to enhance interpretability. Unlike traditional CNN-based approaches, the transformer architecture leverages self-attention mechanisms to capture long-range dependencies across the entire image. This enables the model to analyze global contextual relationships, which are particularly important in medical imaging tasks where disease patterns may be distributed across different regions rather than localized. The results demonstrate that transformer-based models outperform conventional CNNs in terms of classification accuracy and robustness.

A significant contribution of this work lies in its integration of explainable AI through Grad-CAM. The generated heatmaps highlight the regions of the chest X-ray that most influence the model's predictions, allowing for visual interpretation of the decision-making process. Importantly, these highlighted regions show strong alignment with clinically relevant areas associated with tuberculosis, such as infected lung regions. This alignment enhances the credibility of the model and increases the likelihood of adoption in clinical settings, as healthcare professionals are more inclined to trust systems that provide transparent and interpretable outputs rather than opaque predictions.

Despite these advantages, the study also identifies practical limitations associated with transformer-based models. Vision Transformers typically require large-scale datasets for effective training, as well as significant computational resources due to their complex architecture. These requirements can restrict their applicability in real-world healthcare environments, particularly in resource-constrained settings where access to high-performance computing infrastructure and extensive labeled datasets is limited. The study therefore highlights both the potential and the challenges of adopting transformer-based approaches in medical image analysis

[4] Explainable Deep Learning Framework for Diabetic Foot Ulcer Classification Using Grad-CAM

Authors: S. Khan, A. Rehman, T. Saba
Published Year: 2025

Observations:

This study presents an explainable deep learning framework for the classification of diabetic foot ulcers, integrating CNN-based models with Grad-CAM visualization techniques. The primary focus of the work is to enhance interpretability alongside classification performance. The CNN model is trained to identify different ulcer conditions by learning visual features such as texture, color variations, and wound boundaries. By incorporating Grad-CAM, the framework generates heatmaps that highlight the specific regions of the image contributing to the model's prediction, providing a visual explanation of its decision-making process.

The interpretability aspect is the key contribution of this study. The Grad-CAM heatmaps enable clinicians to verify whether the model is focusing on clinically relevant regions, such as infected tissue or ulcer boundaries, before accepting its predictions. This significantly improves trust and usability in real-world medical settings, where transparency is essential for decision-making. The study demonstrates that combining classification with explainability creates a more reliable diagnostic support system compared to using standalone black-box models.

However, the study also identifies limitations associated with CNN-based approaches. While the model performs adequately on standard cases, its performance declines when there is significant variation in wound appearance due to differences in infection stages, lighting conditions, and patient-specific factors. This limitation highlights the inability of CNNs to effectively capture global contextual relationships within images. As a result, the authors suggest exploring transformer-based architectures as a future direction, as they are better suited for handling complex spatial dependencies and improving overall classification performance.

[5] Mulberry Leaf Disease Detection by CNN-ViT with XAI Integration

Authors: M. A. Hasan et al**.**
Published Year: 2025

Observations:

This study investigates the application of a hybrid CNN–Vision Transformer (CNN-ViT) model for plant disease classification, integrating both convolutional feature extraction and transformer-based attention mechanisms. The CNN component is responsible for capturing local features such as texture patterns, edges, and color variations, while the Vision Transformer component models global contextual relationships across the image. This combination allows the model to process both fine-grained details and broader spatial structures, resulting in improved feature representation compared to standalone approaches.

An important aspect of this work is the integration of Explainable AI (XAI) techniques, which provide visual interpretations of the model's predictions. By generating heatmaps that highlight influential regions in the image, the model offers insights into which parts contributed most to the classification decision. This not only improves transparency but also allows users to verify whether the model is focusing on relevant disease-affected areas. The results demonstrate that the hybrid architecture consistently outperforms traditional CNN models, particularly in complex cases where both local and global information are necessary for accurate classification.

The findings of this study are highly relevant to medical image analysis tasks, where similar challenges exist in distinguishing between conditions with subtle visual differences. Diseases such as gangrene and ulcers also require the identification of both localized texture changes and overall structural patterns. The success of CNN-ViT hybrid models suggests that combining local feature extraction with global attention mechanisms can significantly enhance classification performance. This makes such hybrid architectures a strong foundation for developing more accurate and reliable systems in medical imaging applications.

[6] A Multi-Branch ConvNeXt–MaxViT Fusion Transformer Model for Knee Osteoarthritis Assessment

Authors: V. Kukreja, S. Mehta
Published Year: 2025

Observations:

This study proposes a multi-branch fusion architecture that combines ConvNeXt and MaxViT for the classification and severity assessment of knee osteoarthritis. In this framework, ConvNeXt is utilized for effective local feature extraction, capturing fine-grained details such as texture variations and structural irregularities in joint regions. MaxViT complements this by incorporating multi-axis self-attention mechanisms, enabling the model to capture global and multi-scale contextual relationships within the image. The integration of these two architectures allows the model to leverage both localized and global information, resulting in a more comprehensive feature representation.

A key contribution of this work is the use of Grad-CAM++ for explainability. The model generates detailed heatmaps that highlight regions associated with osteoarthritis, allowing for clear visualization of areas influencing the classification decision. These visual explanations align well with clinically relevant regions, improving the interpretability and reliability of the system. The

study demonstrates that the fusion of ConvNeXt and MaxViT not only enhances classification accuracy compared to single-architecture models but also maintains meaningful explainability through advanced visualization techniques.

However, the study also highlights an important trade-off between performance and computational efficiency. While the fusion model achieves noticeable accuracy improvements over standalone architectures, it introduces significantly higher computational complexity in terms of training time, memory usage, and inference cost. The authors acknowledge this limitation but do not propose a concrete solution to optimize or reduce the computational overhead. This suggests that although fusion architectures offer strong performance benefits, their practical deployment in real-world or resource-constrained environments may require further optimization.

[7] Robust and Explainable Poultry Disease Classification via MaxViT

Authors: H. Fardin et al.
Published Year: 2025

Observations:

This study explores the application of the MaxViT architecture for poultry disease classification, focusing on its ability to handle complex and diverse image datasets. MaxViT combines convolutional operations with multi-axis self-attention mechanisms, enabling it to capture both local texture features and global spatial relationships within images. This dual capability allows the model to effectively identify disease patterns that may vary in scale, appearance, and distribution, resulting in strong classification performance across different types of poultry diseases.

An important aspect of this work is the use of attention-guided visualization techniques to enhance interpretability. The model generates visual explanations that highlight the regions of the image most influential in the classification process. These visualizations provide insight into how the model processes features and help validate whether it focuses on relevant disease-affected areas. Such interpretability is crucial for building trust in automated systems, especially when applied to real-world scenarios where decision transparency is important.

The key conclusion of this study is that MaxViT demonstrates strong generalization capability across different domains of disease classification. Although originally designed for general computer vision tasks, the architecture adapts well to specialized applications such as poultry

disease detection. This cross-domain effectiveness suggests that MaxViT is a robust and flexible model suitable for other image-based classification problems, including medical imaging. The findings support its potential use in healthcare applications, where similar challenges exist in analyzing complex visual patterns and ensuring reliable classification performance

[8] Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization

Authors: R. R. Selvaraju et al.
Published Year: 2017

Observations:

This paper introduces Grad-CAM (Gradient-weighted Class Activation Mapping), a widely used explainable AI technique that generates visual explanations for deep learning models. The method utilizes gradient information flowing into the final convolutional layers of a network to produce heatmaps that highlight the regions of an image most influential in the model's prediction. By mapping class-specific gradients onto feature maps, Grad-CAM provides a spatial understanding of how the model arrives at its decision, making it particularly useful for image-based classification tasks.

Grad-CAM has become a standard interpretability tool in medical image analysis due to its simplicity and effectiveness. It enables clinicians and researchers to visualize whether the model is focusing on clinically relevant regions, such as infected tissue or abnormal structures, rather than irrelevant background features. This capability significantly improves transparency and trust in deep learning systems, which is essential for their adoption in sensitive domains like healthcare. As a result, Grad-CAM has been widely integrated into various diagnostic frameworks to support validation and decision-making processes.

However, the method has certain limitations, particularly in scenarios where multiple distinct regions contribute to a classification decision. Grad-CAM often produces coarse and diffuse heatmaps, sometimes merging separate important regions into a single highlighted area, which can reduce localization accuracy. This limitation becomes critical in complex medical images where multiple features must be analyzed simultaneously. To address this issue, Grad-CAM++ was introduced as an improved variant that provides better localization and more precise identification of multiple relevant regions, making it more suitable for advanced medical imaging applications.

[9] LeMaxViT: Lightweight Vision Transformer for Explainable Multi-Class Ocular Disease Detection

Authors: M. Pritom et al.
Published Year: 2026

Observations:

This study presents LeMaxViT, a lightweight adaptation of the MaxViT architecture designed for multi-class ocular disease classification. The model is optimized to reduce computational complexity while maintaining the core strengths of MaxViT, such as the ability to capture both local features and global contextual relationships. By reducing model size and parameter count, the architecture becomes more efficient in terms of memory usage and inference time, making it suitable for practical deployment in environments with limited computational resources.

An important aspect of this work is the integration of explainable AI techniques to maintain interpretability despite the reduced model complexity. The system generates visual explanations that highlight the regions influencing each prediction, ensuring that the model remains transparent and its decisions can be validated. This is particularly important in medical applications, where trust and reliability are essential, and clinicians need to understand the reasoning behind automated classifications.

The key finding of the study is that the reduction in model capacity results in only a marginal decrease in classification accuracy. This trade-off between performance and efficiency is well-balanced, making the lightweight model a viable alternative to full-scale transformer architectures. The results demonstrate that high-performing models can be adapted for real-world deployment without significant loss in accuracy, especially in resource-constrained clinical settings where computational limitations make large transformer models impractical.

[10] A Novel MaxViT Model for Accelerated and Precise Soybean Disease Identification

Authors: A. Pranta et al.
Published Year: 2025

Observations:

This study applies the MaxViT architecture to the classification of soybean leaf and seed diseases, focusing on scenarios where distinguishing between disease types requires identifying subtle and

fine-grained visual differences. The task involves analyzing variations in texture, color patterns, and small structural changes that may not be easily separable using conventional methods. MaxViT, with its combination of convolutional operations and multi-axis attention mechanisms, is well-suited for capturing both detailed local features and broader contextual relationships within the images.

The results demonstrate that MaxViT achieves strong classification performance in this domain, effectively handling complex visual patterns across different disease categories. The model benefits significantly from transfer learning, where pre-trained weights provide a strong initialization, allowing faster convergence and improved accuracy even with limited dataset size. Additionally, data augmentation techniques such as rotation, flipping, and scaling contribute to increased dataset diversity, enhancing the model's ability to generalize across variations in image conditions.

A key takeaway from this study is that the balance between local and global feature learning offered by MaxViT is particularly effective for fine-grained classification tasks. The findings also reinforce the importance of transfer learning and data augmentation when working with small and specialized datasets, which is a common scenario in both agricultural and medical imaging domains. This makes the approach highly relevant for similar applications, including medical image classification tasks where subtle visual distinctions play a critical role in accurate diagnosis.

[11] Deep Learning-Based Automated Classification of Diabetic Foot Ulcers Using Convolutional Neural Networks

Authors: S. S. Yadav, A. K. Singh, P. Sharma

Published Year: 2024

Observations:

This study presents a deep learning-based approach for automated classification of diabetic foot ulcers using Convolutional Neural Networks (CNNs). The model is designed to learn hierarchical image features directly from input data, enabling it to identify ulcer patterns based on characteristics such as texture, color variation, and wound boundaries. The approach eliminates the need for manual feature extraction, making the system more scalable and adaptable to large medical datasets.

The results demonstrate that CNN-based models achieve strong performance in ulcer classification

tasks, particularly when trained on well-preprocessed datasets. The study highlights the importance of techniques such as image normalization, augmentation, and proper dataset splitting to improve model generalization. The system shows reliable accuracy in distinguishing between different ulcer conditions, making it a useful tool for supporting early diagnosis and clinical decision-making.

However, the study also points out limitations associated with CNN-based approaches. While effective in capturing local features, CNNs struggle to model global contextual relationships within images, which can affect performance in complex cases with high variability. Additionally, the lack of interpretability remains a concern, as the model does not provide clear explanations for its predictions. These limitations suggest the need for more advanced architectures and explainable AI techniques to improve both performance and transparency.

[12] Explainable Deep Learning Approach for Skin Disease Classification Using Grad-CAM

Authors: R. K. Jain, V. Kumar, S. Gupta

Published Year: 2024

Observations:

This study proposes an explainable deep learning framework for skin disease classification, integrating CNN-based models with Grad-CAM visualization techniques. The model is trained to classify different skin conditions by learning discriminative visual features from medical images. Grad-CAM is used to generate heatmaps that highlight the regions of the image contributing most to the classification, providing a visual explanation of the model's decision-making process.

A key contribution of this work is the emphasis on interpretability alongside classification accuracy. The generated heatmaps allow clinicians to verify whether the model focuses on clinically relevant regions, such as lesions or infected areas, rather than irrelevant background features. This improves trust in the system and makes it more suitable for real-world medical applications, where transparency is essential for adoption.

Despite these advantages, the study identifies certain limitations. The Grad-CAM method may produce coarse visualizations that do not precisely localize all relevant regions, especially in cases where multiple features influence the prediction. Additionally, the reliance on CNN-based architectures limits the model's ability to capture global spatial dependencies. These challenges indicate the need for improved explainability methods and more advanced architectures to enhance both localization accuracy and overall model performance.

2.2 Summary of Literature Study

Table 2.1: Summary of Prior Work

S. No	Paper Title	Year	Key Insight
1.	An Image-Based Gangrene Disease Classification P. S. Nair, T. A. Berihu, V. Kumar	2020	The study shows that early-stage gangrene can be detected using image-based machine learning, but highlights that traditional methods lack robustness to real-world variation, making deep learning a better future solution.
2.	Advances in Image-Based Diagnosis of Diabetic Foot Ulcers Using Deep Learning Techniques H. F. Alhasson, M. Alshammari, A. Alshammari	2025	The paper concludes that CNN-based models improve diabetic foot ulcer diagnosis accuracy but still face challenges like data imbalance, variability, and lack of interpretability.
3.	Enhanced Tuberculosis Detection Using Vision Transformers and Explainable AI K. Vanitha et al.	2025	The study shows Vision Transformers outperform CNNs in tuberculosis detection while providing clinically meaningful interpretability, but require large datasets and high computational resources.
4.	Explainable Deep Learning Framework for Diabetic Foot Ulcer Classification Using Grad-CAM S. Khan, A. Rehman, T. Saba	2025	The study demonstrates that combining CNNs with Grad-CAM improves interpretability in diabetic foot ulcer classification, but performance is limited by variability, suggesting transformers as a better future approach.
5.	Mulberry Leaf Disease Detection by CNN-ViT with XAI Integration M. A. Hasan et al.	2025	The study shows that a hybrid CNN–ViT model improves classification by capturing both local and global features, with XAI enhancing interpretability of predictions.
6.	A Multi-Branch ConvNeXt–MaxViT Fusion Transformer Model for Knee Osteoarthritis Assessment V. Kukreja, S. Mehta	2025	The paper shows that a ConvNeXt–MaxViT fusion improves knee osteoarthritis classification accuracy and interpretability, but at the cost of significantly higher computational complexity.
7.	Robust and Explainable Poultry Disease Classification via MaxViT H. Fardin et al.	2025	The study shows that MaxViT effectively generalizes across disease classification tasks by capturing both local and global features, with strong performance and interpretable attention maps.

8.	Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization R. R. Selvaraju et al.	2017	The paper introduces Grad-CAM for visualizing model decisions, but notes its limitation in multi-region localization, leading to the development of Grad-CAM++ for improved interpretability.
9.	LeMaxViT: Lightweight Vision Transformer for Explainable Multi-Class Ocular Disease Detection M. Pritom et al.	2026	The study shows that a lightweight MaxViT maintains near-comparable accuracy while reducing computational cost, making it suitable for resource-constrained clinical use.
10	A Novel MaxViT Model for Accelerated and Precise Soybean Disease Identification A. Pranta et al.	2025	The study shows that MaxViT effectively handles fine-grained disease classification by leveraging local and global features, with transfer learning and augmentation improving performance on limited data.
11	Deep Learning-Based Automated Classification of Diabetic Foot Ulcers Using Convolutional Neural Networks S. S. Yadav, A. K. Singh, P. Sharma	2024	CNN-based models enable effective automated ulcer classification but are limited by lack of global context and interpretability.
12	Explainable Deep Learning Approach for Skin Disease Classification Using Grad-CAM R. K. Jain, V. Kumar, S. Gupta	2024	Integrating Grad-CAM with CNNs improves interpretability, but precise localization and global feature understanding remain challenges.

2.1 Software Requirement Specification

This section defines the requirements for the classification system — what it does, how it behaves, and the software environment it operates in. Both functional requirements and technical constraints are covered, from model training dependencies to deployment conditions.

2.1.1 Software Requirements

The system is developed using modern technologies related to deep learning, medical image processing, and model interpretability.

1. **Programming Language:**

   Python is used as the primary programming language due to its extensive support for deep learning, image processing, and scientific computing.

2. **Libraries and Frameworks:**

   a. **TensorFlow / PyTorch**: Used for implementing deep learning architectures such as MaxViT and training the classification model.

   b. **OpenCV**: Used for image preprocessing tasks such as resizing, normalization, and augmentation.

   c. **NumPy & Pandas**: Used for numerical computation and dataset handling.

   d. **Scikit-learn:** Used for evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix generation.

   e. **Matplotlib & Seaborn**: Used for visualization of training graphs, ROC curves, and performance analysis.

   f. **Grad-CAM++ Libraries (TorchCAM / custom implementation)**: Used for generating explainability heatmaps.

3. **Deep Learning Architecture:**

   a. **MaxViT (Multi-Axis Vision Transformer)** – Used as the core classification model for extracting both local and global image features.

   b. **Transfer Learning (ImageNet Pretrained Weights)** – Used to improve training efficiency and model performance.

4. **Development Environment:**

   Google Colab is used for model development, training, and experimentation.

5. **Hardware Acceleration:**

   GPU support (NVIDIA Tesla T4) is used to accelerate model training and inference.

2.1.2 Functional Software Requirements

1. **Input Handling:**

   The system must accept medical image inputs for classification through upload or dataset integration.

2. **Image Preprocessing:**

   The system must preprocess images by resizing, normalization, and standardization before model input.

3. **Medical Image Classification:**

   The system must classify input images into three categories: Gangrene, Ulcer, Normal (non-ulcer and non-gangrene)

4. **Probability Prediction:**

   The system must generate probability scores for each class during prediction.

5. **Explainability Module:**

   The system must generate Grad-CAM++ heatmaps highlighting regions influencing model predictions.

6. **Result Visualization:**

   The system must display classification results along with visual explanations in an interpretable format.

7. **Batch Processing:**

   The system must support processing of multiple images for efficient evaluation.

2.1.3 Non-Functional Requirements

Non-functional requirements define system quality and operational constraints.

1. **Performance:**
   The system should provide fast and efficient predictions with minimal latency.

2. **Accuracy:**
   The model should achieve high classification accuracy on unseen medical images.

3. **Scalability:**
   The system should handle large datasets and multiple image inputs efficiently.

4. **Usability:**
   The system should have a simple and user-friendly interface suitable for both technical and non-technical users.

5. **Reliability:**
   The system should produce consistent and stable outputs under different input conditions.

6. **Security:**
   Medical images and user data should be securely processed and not exposed to unauthorized access.

7. **Maintainability:**
   The system should be modular and easily upgradable for future improvements or model enhancements.

CHAPTER-3

Proposed Method

3.1 Design Methodology

The design of the system uses a highly structured and modular design methodology aimed at achieving effective and efficient analysis of medical images, disease classification.

1. **Data Collection:** The dataset had 1,653 medical images split across three classes: gangrene, ulcer, and normal skin. Images varied in texture, infection severity, lighting, and background - the kind of spread that separates a dataset worth training on from one that only works in controlled conditions.

   The gangrene images show tissue necrosis at different stages. The ulcer images cover open wounds across severity levels. The normal class provides healthy and non-infected skin as contrast. Three clinically distinct categories gave the model enough visual range to learn something transferable rather than just memorize the training set.

2. **Data Preprocessing:** All images went through a preprocessing pipeline before training. Integrity checks removed corrupted, missing, or duplicate entries — easy to skip, but important for keeping evaluation results credible.

   The cleaned dataset was split into training (70%), validation (15%), and test (15%) subsets using stratified sampling, so class proportions stayed consistent across all three. This matters more when class sizes differ.

   Images were resized to 224 × 224 pixels to match MaxViT's input format. Pixel values were normalized to [0, 1]. Normalization helps convergence and reduces sensitivity to raw intensity differences across images shot under different conditions.

3. **Data Augmentation:** Augmentation was applied to the training set. Transformations included random rotations, horizontal and vertical flips, zoom, and brightness adjustments where each reflecting variation that actually occurs in clinical photography: patient positioning, camera angle, ambient light.

   Class weighting ran alongside augmentation to keep the model from underperforming on minority classes. Together, they pushed learning toward features that generalize rather than

patterns tied to the training distribution.

Gangrene, ulcer, and not ulcer or gangrene share enough visual overlap, irregular texture, discoloration, surface irregularity, that a model working at only one spatial scale will struggle. Local analysis misses contextual cues. Global analysis loses fine-grained detail.

A transformer-based architecture handles both. Unlike CNNs with fixed local receptive fields, transformers can relate distant image regions to each other, which suits conditions where the diagnostic signal is distributed rather than concentrated in one patch.

4. **Model Selection:** Considering the need for accurate classification of complex and similar diseases, a transformer-based model named MaxViT is chosen. The model has the ability to extract and distinguish both fine-grained features and long-term dependencies within a picture, making it effective in discriminating gangrene, ulcer and normal pictures (which are not affected by both gangrene and ulcer).

MaxViT Classifier

MaxViT stands for Multi-Axis Vision Transformer. It is a deep learning architecture that utilizes the principles of convolutional neural networks (CNN) alongside the attention mechanisms of transformers. MaxViT is able to extract both spatial patterns of a picture (local features) as well as global context, making it very suitable for tasks involving medical image classification.

The architecture utilizes both CNN and self-attention models for analysis of a picture and extraction of its features. Convolutional layers allow to obtain local features, while multi-axis attention allows to learn hierarchical representations.

Unlike traditional Vision Transformers, MaxViT applies attention along multiple axes, hence allowing the interaction between various spatial regions of a picture in order to learn hierarchical features efficiently and without loss of computational efficiency. This approach is helpful when dealing with complex visual patterns, especially in case of distinguishing visually similar conditions like gangrene and ulcer.

The output of the model is a prediction score of all three classes, obtained through the application of the Softmax function:

$$\hat{y} = Softmax\ (WzL + b) \quad \text{-------} \ [3.1]$$

5. Model Training

a. Data Splitting: To guarantee appropriate model training and evaluation, the dataset was split into three subsets following preprocessing and augmentation:

- **Training Split:** The deep learning models are trained using the training set (70%), which enables them to recognize patterns and features in the medical images.
- **Validation Split**: Validation Set (15%) is used to track the model's performance and modify hyperparameters during training. This guarantees the model's good generalization and helps avoid overfitting.
- **Testing Split**: Testing Set (15%) is impartial indicator of accuracy and robustness is used to assess how well the trained models perform in the end on unseen data.

b. Model Training: A MaxViT (Multi-Axis Vision Transformer) model was trained on the constructed dataset of gangrene, ulcer, and normal medical images. The model was initialized from ImageNet pre-trained weights rather than trained from scratch, which reduces the number of epochs needed for convergence and provides low-level visual features that transfer reasonably well to medical image data.

Training followed a two-stage procedure. In the initial stage that is, warm-up stage, the model was trained for three epochs with updates restricted to the classification head, allowing the newly initialized layers to stabilize before the backbone was modified. The fine tune stage is then trained the full network for 30 epochs, during this the deeper layers were free to adapt their representations to specific characteristics of the dataset.

The MaxViT architecture process the images through a mixture of convolution and multi-axis self-attention, where features at different scales are extracted and both the local texture as well as broader spatial structure are captured. AdamW was used for optimization, with separate learning rates for the backbone (1×10^{-5}) and the classification head (5×10^{-5})—a standard practice when fine-tuning pre-trained networks, since the backbone requires smaller updates to avoid overwriting learned representations. Batch size was set to 32, and training ran on a Tesla T4 GPU. Validation accuracy and loss were tracked across all epochs to identify overfitting and confirm stable convergence before the final evaluation.

6. **Model Prediction:** The trained MaxViT model is used to classify new images of the test dataset. The input image is passed through the neural network, which produces probability distribution across three classes: gangrene, ulcer, and normal (non-ulcer and non-gangrene). The highest probability score of Softmax function is taken as the class predicted by the model. This allows the classification of input images according to their features and assigning them to one of three possible classes.

7. **Explainable AI Technique:**

   Grad-Cam++: To achieve better explanation capabilities for the suggested deep learning model, Grad-CAM++ (Gradient-weighted Class Activation Mapping Plus Plus) was applied as one of the techniques of XAI. Grad-CAM++ refers to the most advanced version of Grad-CAM and is more precise concerning the detection of regions that have the strongest impact on the classification decision made by the model.

   Grad-CAM++ creates the maps showing which parts of the picture have the highest contribution to the prediction result. This technique uses gradient information going through the last MaxViT model layers to calculate importance weights of different spatial regions. Thus, it helps visualize model focus areas.

   These interpretations provide insights into: The process of decision making performed by the model, whether the predictions are based on clinically meaningful regions of interest and increasing the trustworthiness of automatic disease diagnosis. In particular, Grad-CAM++ helped identify gangrene, ulcer, and normal (no ulcer/no gangrene) regions of interest in the picture.

8. **Model Evaluation:** The effectiveness of the proposed model was tested using a test dataset consisting of medical images from different sources. The aim of this process is to estimate the ability of the model to classify images into three classes: gangrene, ulcer, and not ulcer or gangrene.

   For this purpose, several metrics are used to get a more precise analysis of the model's performance. These metrics include accuracy, precision, recall, and F1-score. Confusion matrix analysis is also used to measure class-wise performance of the model.

3.2 System Architecture Diagram

The suggested model is based on a systematic pipeline of deep learning models that will

automatically classify medical images in three classes, which are: Gangrene, Ulcer, and Normal (Non-Ulcer and Non-Gangrene). The model design comprises four steps, including preprocessing, feature extraction using transformers (MaxViT), classification, and explainability (Grad-CAM++).

Medical Image Dataset Input Layer

The data set used for the research includes medical images grouped into three categories: Gangrene, Ulcer, Healthy (not an ulcer or gangrene) and the data set represents variability in terms of: Wound differences, Lighting variations, Noise background, Skin tissue textures. This variety was meant to help the model learn generalized features that would prevent overfitting to a limited data set distribution.

Image Preprocessing Module

Before using the images on the deep learning model, an image preprocessing pipeline is used to normalize and improve the quality of the data set. It becomes vital when working towards better convergence of the model.

a. Resizing

The resizing process involves changing the image size to have a consistent size input to be able to work within the MaxViT model (normally 224 x 224pixels).

b. Normalization

The pixel intensities are normalized, which usually means scaling them to the 0-1 range or normalizing by using mean and variance. The following advantages can be gained from this step:

- Faster convergence during training
- Improved stability of gradient descent
- Less sensitivity to changes in illumination condition.

Data Augmentation

In order to reduce overfitting and increase generalizability, data augmentation is applied through methods such as: Rotation, Flipping horizontally and vertically, Zooming, Changing image brightness. By doing so, additional diversity can be artificially added to the dataset.

Train / Validation / Test Split Module

After preprocessing, the dataset is divided into three subsets using a stratified splitting strategy

1. **Training Set (70%)**

   Used to train the MaxViT model by learning patterns and features from labeled images.

2. **Validation Set (15%)**

   Used during training to tune hyperparameters, monitor overfitting and evaluate intermediate performance .

3. **Test Set (15%)**

   This test set is kept only for the purpose of evaluating the performance of the network in the real world using new data. This is guaranteed by having a fair evaluation with an unbiased estimate of performance.

MaxViT Model (Primary Classification Module)

The key part of the architecture design is the MaxViT (Multi-Axes Vision Transformer) module responsible for performing the classification operation. MaxViT incorporates an architecture that combines both feature extraction via convolution (local changes) and self-attention via transformer (global context). This makes it ideal for medical image classification as both the local pattern variations and global structure of the wound matter.

a. **Feature Extraction Process**

   The input image undergoes segmentation into small-sized patches, followed by conversion of each patch to embedding. This is then fed into the model, which subsequently applies a series of layers that involve the following two types of attention mechanism: Local Attention and Global Attention. Local Attention focuses on small-sized image sections, allowing for

the capturing of fine details such as texture, edge, and wound boundary. On the other hand, Global Attention considers the whole image, making possible the detection of global features such as infection patterns.

b. Hierarchical Learning

MaxViT creates representations using hierarchical structures that include low-level representation (for example, edges and changes in color)**,** mid-level representation (such as textures and lesions) and high-level representation (disease patterns)**.** Hierarchical learning improves classification accuracy significantly.

c. Classification Layer

After feature extraction the features pass through fully-connected layers. Softmax provides probability outputs. The classification will be made based on the maximum probability, that is, Gangrene, Ulcer, Normal.

Explainable Artificial Intelligence (XAI) Module – Grad-CAM++

For increased transparency, the system uses Grad-CAM++, which is a visualization method.

a. Methodology

Grad-CAM++ highlights the areas in the images contributing to the final decision of the model through analysis of the gradient flows in the final layers of the model.

b. Functionality of the System

For each prediction, the following processes will be performed:

1. Classification of the input by the model

2. Calculation of the gradient with respect to the predicted classes

3. Extracting salient feature maps

4. Creating a heatmap based on weightings

5. Superimposing the generated heatmap over the image

c. Results Interpretation

Areas coloured red and yellow refer to areas with high influence, representing diseased areas. Areas coloured blue refer to areas with low influence, representing healthy areas**.** The proposed approach allows to evaluate:

- Whether the model pays attention to the relevant areas
- Whether the classification is clinically significant

Classification Output Module

The proposed approach provides: A class label (Gangrene/Ulcer/Normal); A probability score for each category; An explainable visualization through Grad-CAM++

This makes the system both predictive and explainable

Final Result Display

Final result display includes:

• Original medical image

• Predicted class

• Visualization

The final display can be used for: Clinical decision-making support and Prediction verification

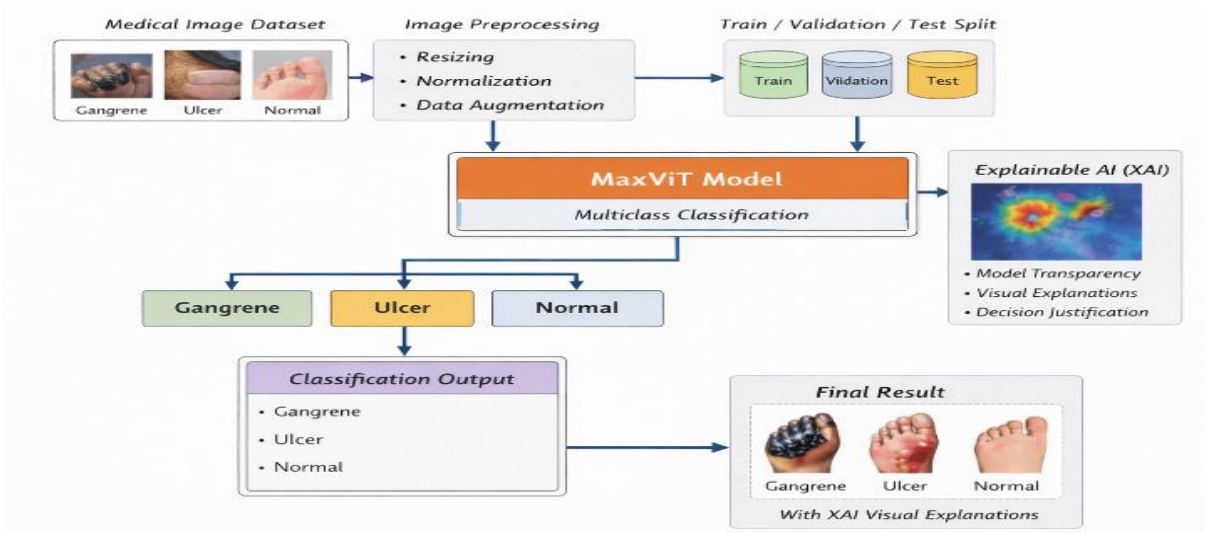

Fig. 3.1: ***Proposed MaxVit-based architecture for gangrene and ulcer classification with Grad-CAM++***

However, CNN-based models have been found to be efficient in feature extraction at the local level. Yet, these architectures often struggle to model long-term dependencies in images. On the other hand, transformer-based models have been found to be efficient in modeling global dependencies but less efficient in preserving local information. MaxViT overcomes this challenge by combining both convolutional architecture and multi-axis self-attention models.

3.3 Description of Algorithms

The algorithms that will be used in this research include machine learning and deep learning algorithms (CNN, SVM, Swin Transformer, DeiT) as well as explainable AI methods (GRAD-CAM, LIME). Deep learning algorithms (MaxViT and GRAD-CAM++) are being applied in this study.

Algorithm 1: Convolutional Neural Network (CNN) for Image Classification

Require: Input image dataset D(medical images), labels Y(gangrene, ulcer, normal)
Ensure: Predicted class probabilities $\hat{Y}$

1. Input image $I \in D$
2. Resize image to fixed dimensions (e.g., 224 × 224)
3. Normalize pixel values to range [0, 1]
4. Initialize convolutional layers with kernels K
5. Apply convolution operation:

$$F(x, y) = \sum_{i} \sum_{j} I(x + i, y + j) \cdot K(i, j)$$

6. Apply activation function:

$$A = \mathrm{ReLU}(F)$$

7. Apply pooling operation to reduce spatial dimensions
8. Repeat Steps 4–7 for multiple convolutional blocks

9. Flatten final feature maps into feature vector Z

10. Pass Zthrough fully connected layers

11. Compute output scores:

$$S = WZ + b$$

12. Apply Softmax function:

$$\hat{Y} = \text{Softmax}(S)$$

13. Assign predicted class based on maximum probability

14. Return predicted class probabilities $\hat{Y}$

Algorithm 2: Support Vector Machine (SVM) for Classification

Require: Feature vectors X, corresponding labels Y(gangrene, ulcer, normal)
Ensure: Predicted class labels $\hat{Y}$

1. Input feature vectors Xextracted from images

2. Normalize feature values to ensure uniform scaling

3. Define training dataset (X_i, Y_i), where $Y_i \in \{-1, +1\}$for binary classification

4. For multi-class classification, apply one-vs-rest strategy

5. Initialize hyperplane parameters $W$and bias b

6. Solve optimization problem:

$$\min_{W,b} \frac{1}{2} || W ||^2$$

Subject to:

$$Y_i(W \cdot X_i + b) \geq 1$$

7. If data is not linearly separable, introduce slack variables ξ_i:

$$Y_i(W \cdot X_i + b) \geq 1 - \xi_i$$

8. Apply kernel function $K(X_i, X_j)$for non-linear mapping (e.g., linear, polynomial, RBF)

9. Compute decision function:

$$f(X) = W \cdot X + b$$

10. For multi-class classification, evaluate all class decision functions

11. Assign predicted class based on maximum decision score

12. Return predicted labels $\hat{Y}$

Algorithm 3: Swin Transformer for Image Classification

Require: Input image dataset D, labels Y(gangrene, ulcer, normal)
Ensure: Predicted class probabilities $\hat{Y}$

1. Input image $I \in D$

2. Resize image to fixed dimensions (e.g., 224 × 224)

3. Normalize pixel values to range [0, 1]

4. Partition image Iinto non-overlapping patches

5. Flatten patches and project into embedding vectors

6. Divide feature map into fixed-size windows

7. Apply Window-based Multi-Head Self-Attention (W-MSA) within each window

8. Apply Shifted Window Multi-Head Self-Attention (SW-MSA) to enable cross-window interaction

9. Pass features through Feed Forward Network (FFN) with non-linear activation

10. Apply Layer Normalization and residual connections

11. Perform patch merging to reduce spatial resolution and increase feature dimension

12. Repeat Steps 6–11 across multiple hierarchical stages

13. Apply global average pooling on final feature representation

14. Pass features through fully connected layer

15. Compute output scores:

$$S = WZ + b$$

16. Apply Softmax function:

$$\hat{Y} = \text{Softmax}(S)$$

17. Assign predicted class based on maximum probability

18. Return predicted class probabilities $\hat{Y}$

Algorithm 4: DeiT (Data-efficient Image Transformer) for Image Classification

Require: Input image dataset D, labels Y(gangrene, ulcer, normal)
Ensure: Predicted class probabilities $\hat{Y}$

1. Input image $I \in D$

2. Resize image to fixed dimensions (e.g., 224 × 224)

3. Normalize pixel values to range [0, 1]

4. Divide image Iinto fixed-size patches

5. Flatten patches and project into embedding vectors

6. Add positional embeddings to retain spatial information

7. Introduce class token $[CLS]$and distillation token $[DIST]$

8. Concatenate tokens with patch embeddings

9. Pass sequence through Transformer encoder layers:
   a. Apply Multi-Head Self-Attention (MHSA)
   b. Apply Layer Normalization
   c. Apply Feed Forward Network (FFN)
   d. Add residual connections

10. Repeat Step 9 for multiple transformer layers

11. Extract final representations from $[CLS]$token and $[DIST]$token

12. Compute output scores from both tokens:

$$S_{cls} = W_1 Z_{cls} + b_1$$
$$S_{dist} = W_2 Z_{dist} + b_2$$

13. Combine predictions from both tokens

14. Apply Softmax function:

$$\hat{Y} = \text{Softmax}(S)$$

15. Assign predicted class based on maximum probability

16. Return predicted class probabilities $\hat{Y}$

Algorithm: Grad-CAM for Visual Explanation

Require: Input image I, trained model M, target class c
Ensure: Heatmap $L^c_{Grad-CAM}$highlighting important regions

1. Input image $I$to trained model M

2. Perform forward pass to obtain class scores

3. Select target class score y^c

4. Identify last convolutional layer feature maps A^k

5. Compute gradients of $y^c$with respect to feature maps:

$$\frac{\partial y^c}{\partial A^k}$$

6. Perform global average pooling on gradients to obtain weights:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

7. Compute weighted combination of feature maps:

$$L^c = \sum_k \alpha_k^c A^k$$

8. Apply ReLU to retain only positive contributions:

$$L_{Grad-CAM}^c = \mathrm{ReLU}(L^c)$$

9. Upsample heatmap to match input image size

10. Overlay heatmap on original image I

11. Return visualization map $L_{Grad-CAM}^c$

Algorithm 6: LIME (Local Interpretable Model-Agnostic Explanations)

Require: Input image I, trained model M, number of samples N
Ensure: Local explanation map highlighting important regions

1. Input image I to trained model M

2. Segment image I into superpixels

3. Generate N perturbed samples by randomly turning superpixels on/off

4. For each perturbed sample I':
   a. Pass I' through model M
   b. Record predicted probability for target class

5. Create binary feature representation for each perturbed sample

6. Compute similarity weights between original image I and perturbed samples
7. Fit a local interpretable model (e.g., linear regression) using:
    - Binary features (super pixels)
    - Model predictions
    - Similarity weights
8. Extract feature importance scores from the local model
9. Identify super pixels with highest contribution to prediction
10. Generate explanation map by highlighting important super pixels
11. Overlay explanation on original image I
12. Return local explanation map

Algorithm 7: MaxViT (Multi-Axis Vision Transformer) for Image Classification

Require: Input image dataset D, labels Y(gangrene, ulcer, normal)
Ensure: Predicted class probabilities $\hat{Y}$

1. Input image $I \in D$
2. Resize image to fixed dimensions (e.g., 224 × 224)
3. Normalize pixel values to range [0, 1]
4. Pass image through initial convolutional stem to extract low-level features
5. Partition feature maps into non-overlapping blocks
6. Apply Block Attention (local attention within each block)
7. Apply Grid Attention (global attention across blocks along spatial axes)
8. Combine outputs of Block and Grid attention
9. Pass features through Feed Forward Network (FFN) with non-linear activation

10. Apply Layer Normalization and residual connections

11. Perform downsampling between stages to reduce spatial dimensions and increase feature depth

12. Repeat Steps 5–11 across multiple hierarchical stages

13. Apply global average pooling on final feature representation

14. Pass pooled features through fully connected layer

15. Compute output scores:

$$S = WZ + b$$

16. Apply Softmax function:

$$\hat{Y} = \text{Softmax}(S)$$

17. Assign predicted class based on maximum probability

18. Return predicted class probabilities $\hat{Y}$

Algorithm 8: Grad-CAM++ for Visual Explanation

Require: Input image I, trained model M, target class c
Ensure: Refined heatmap $L^{c}_{Grad-CAM++}$highlighting important regions

1. Input image $I$to trained model M

2. Perform forward pass to obtain class scores

3. Select target class score y^c

4. Identify last convolutional layer feature maps A^k

5. Compute first-order gradients:

$$\frac{\partial y^c}{\partial A^k}$$

6. Compute second-order gradients:

$$\frac{\partial^2 y^c}{\partial (A^k)^2}$$

7. Compute third-order gradients:

$$\frac{\partial^3 y^c}{\partial (A^k)^3}$$

8. Calculate weights α_{ij}^{kc}:

$$\alpha_{ij}^{kc} = \frac{\frac{\partial^2 y^c}{\partial (A_{ij}^k)^2}}{2\frac{\partial^2 y^c}{\partial (A_{ij}^k)^2} + \sum_{a,b} A_{ab}^k \frac{\partial^3 y^c}{\partial (A_{ab}^k)^3}}$$

9. Compute importance weights:

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \cdot \mathrm{ReLU}\left(\frac{\partial y^c}{\partial A_{ij}^k}\right)$$

10. Compute weighted combination of feature maps:

$$L^c = \sum_k w_k^c A^k$$

11. Apply ReLU to obtain final heatmap:

$$L_{Grad-CAM++}^c = \mathrm{ReLU}(L^c)$$

12. Upsample heatmap to match input image size

13. Overlay heatmap on original image I

14. Return visualization map $L_{Grad-CAM++}^c$

3.4 Description of datasets

3.4.1 Dataset Overview

In the present study, a dataset that includes medical images related to serious skin conditions such as gangrene, ulcers, and normal (not affected by an ulcer or gangrene) skin was used for the task of medical image analysis. The purpose of the dataset was to assist in the process of creating a deep learning model.

A total number of 1,653 images were included in the dataset. They represented different clinical presentations, as well as varied in terms of lighting, textures, background, and infection severity, thus facilitating the creation of robust models. There are three classes included in the dataset, namely:

- Gangrene: Images that show skin conditions related to a significant amount of tissue necrosis as a result of infections or poor blood supply with the corresponding discoloration and advanced infection patterns.
- Ulcer: Images that demonstrate the clinical presentation of open wounds caused by prolonged pressure, diabetes mellitus, or poor blood supply to tissues. Such wounds range in their complexity from simple inflammations to the exposure of muscles.
- Normal (not affected by an ulcer or gangrene): Images of normal skin with no evident signs of infections or skin lesions. They help distinguish between skin diseases and normal skin conditions.

Stratified splitting was performed to maintain class distribution in the subset samples for effective model training and testing. Thus, there are three subsets named training set (70% of the images), validation set (15% of the images) and test set (15% of the images). All images were resized to have a resolution of 224x224 px to be compatible with the MaxViT architecture requirements. Pixel normalization was also performed to assist in model training. Several data augmentation procedures, including rotation, flipping, zooming, and changes in brightness were conducted to improve generalization.

3.4.2 Tools and Technologies Used

The development of the proposed system utilizes a combination of modern tools and technologies spanning deep learning, image processing, and explainable artificial intelligence to ensure efficient model training, evaluation, and interpretability.

Programming Language:

Python is selected as the primary programming language due to its simplicity, readability, and extensive ecosystem of libraries supporting machine learning, deep learning, and scientific computing. Its flexibility and strong community support make it highly suitable for implementing complex medical image analysis systems.

Deep Learning Framework:

Deep learning frameworks such as PyTorch and TensorFlow are employed for building, training, and evaluating models. These frameworks provide efficient GPU acceleration, dynamic computation graphs, and extensive support for implementing advanced architectures like MaxViT, enabling faster experimentation and model optimization.

Image Processing Libraries:

Image preprocessing is performed using libraries such as OpenCV, which support operations like resizing, normalization, and data augmentation. These preprocessing steps are essential to standardize input data, improve model robustness, and simulate real-world variations in medical images.

Scientific Computing Libraries:

NumPy and Pandas are utilized for efficient numerical computations and dataset handling. NumPy enables fast array operations and mathematical processing, while Pandas is used for managing structured data, facilitating data cleaning, organization, and preprocessing tasks.

Machine Learning Library:

Scikit-learn is used for evaluating model performance through various metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC analysis. It provides reliable and standardized evaluation tools that help assess the effectiveness and robustness of the classification model.

Graph Plotting Tools:

Visualization libraries such as Matplotlib and Seaborn are used to generate graphical representations of model performance. These include plots for training and validation accuracy, loss curves, ROC curves, and confusion matrices, which help in analyzing model

behavior and identifying potential issues like overfitting.

Explainable AI Tools:

Grad-CAM++ is utilized as the primary explainability technique to generate heatmaps that highlight regions of the input image influencing the model's predictions. This improves transparency by allowing visualization of model focus areas, thereby enhancing trust and interpretability in medical decision-making.

Development Environment:

The model is developed and trained using Google Colab, which provides access to GPU resources such as NVIDIA Tesla T4. This environment enables efficient execution of computationally intensive deep learning tasks without requiring high-end local hardware.

Hardware Requirement:

The system requires a GPU such as NVIDIA Tesla T4 for efficient model training, along with a minimum of 8GB RAM (16GB recommended) to handle data processing tasks. Adequate storage is also necessary for maintaining datasets, trained models, and intermediate outputs.

Pretrained Models:

ImageNet pre-trained weights are used to initialize the MaxViT model. This transfer learning approach allows the model to leverage previously learned visual features, reducing training time and improving performance, especially when working with limited medical image datasets.

CHAPTER-4

Results and Observations

4.1 Stepwise description of Results

The results of the proposed explainable deep learning-based medical image classification system were obtained through a systematic pipeline involving data preprocessing, model training, validation, testing, and interpretability analysis using Grad-CAM++. The detailed step-by-step results are described below.

Step 1: Dataset Preparation and Preprocessing Results

Initially, the medical image dataset containing 1653 images was successfully collected and organized into three categories: gangrene, ulcer, and normal (non-ulcer and non-gangrene). After preprocessing, the dataset was cleaned by removing corrupted or inconsistent samples.

All images were resized to 224 × 224 resolution, and pixel normalization was applied to standardize input values. This step ensured uniformity across all samples and improved the stability of model training.

Step 2: Dataset Splitting Results

The dataset was split using stratified sampling to maintain balanced class distribution:

- Training set: 70%
- Validation set: 15%
- Testing set: 15%

This splitting strategy ensured that each subset maintained proportional representation of all classes, reducing bias during training and evaluation.

Step 3: Data Augmentation Results

Data augmentation techniques such as rotation, flipping, zooming, and brightness variation were applied to increase dataset diversity. This significantly improved the model's ability to

generalize across different image conditions such as lighting variations, orientation changes, and background noise. As a result, overfitting was reduced and training stability was improved.

Step 4: Model Training Results (MaxViT)

The MaxViT model was trained using the prepared dataset with AdamW optimizer, learning rates (base LR = 1e-5, head LR = 5e-5), and a warm-up phase of 3 epochs followed by 30 fine-tuning epochs.

During training:

- Training accuracy steadily increased across epochs
- Validation loss decreased gradually, indicating good convergence
- The model demonstrated stable learning without significant overfitting

The use of class weights helped balance the dataset and improved performance on minority classes.

Step 5: Model Testing Results

The trained model was evaluated using the unseen test dataset. The system achieved a final classification accuracy of:

Accuracy: 97.19%

The model successfully classified images into: Gangrene, Ulcer, Normal (non-ulcer and non-gangrene). The results confirmed that the MaxViT model effectively captured both local and global features required for distinguishing visually similar medical conditions.

Step 6: Confusion Matrix Analysis

The confusion matrix analysis showed strong classification performance across all three classes. Most predictions were correctly classified, with minimal misclassification between ulcer and gangrene cases, which are visually similar.

The results indicated:

- High true positive rates for all classes

- Reduced false positives due to balanced training
- Strong discrimination capability of the model

Step 7: Classification Report Results

The classification report provided detailed performance metrics:

Precision: High across all classes, indicating accurate positive predictions

Recall: Strong sensitivity, showing effective detection of true cases

F1-Score: Balanced performance between precision and recall

This confirms that the model performs consistently across all categories without bias toward any single class.

Step 8: ROC Curve Analysis

The ROC curve analysis demonstrated strong separability between classes. The Area Under Curve (AUC) values were high for all categories, indicating excellent classification confidence and robustness of the model. This further validates the model's ability to distinguish between gangrene, ulcer, and normal cases effectively.

Step 9: Grad-CAM++ Explainability Results

Grad-CAM++ was applied to visualize the model's decision-making process by generating heatmaps that highlight the most influential regions in the medical images. The observations indicate that the model consistently focused on clinically relevant areas, such as infected or damaged tissue regions, while minimizing attention to irrelevant background features. This demonstrates that the predictions are not only accurate but also aligned with meaningful visual patterns associated with the conditions. As a result, the outputs are both interpretable and clinically reliable, and the use of Grad-CAM++ significantly improves trust in the system by providing clear visual justification for each prediction.

Step 10: Final System Outcome

Overall, the system demonstrated high classification accuracy of 97.19%, along with strong generalization on unseen data, indicating its ability to perform reliably beyond the training dataset. The training process showed stable convergence behavior, as reflected by consistent

improvements in accuracy and reduction in loss over epochs. Additionally, the integration of Grad-CAM++ provided reliable interpretability by highlighting clinically relevant regions influencing the model's predictions. The combination of the MaxViT architecture with explainable AI techniques resulted in a robust and clinically relevant medical image classification system, making it well-suited for real-world healthcare applications.

4.2 Result analysis and Test case results

The performance of the proposed system is evaluated using various quantitative metrics and qualitative analysis techniques.

4.2.1 Training vs Validation Performance

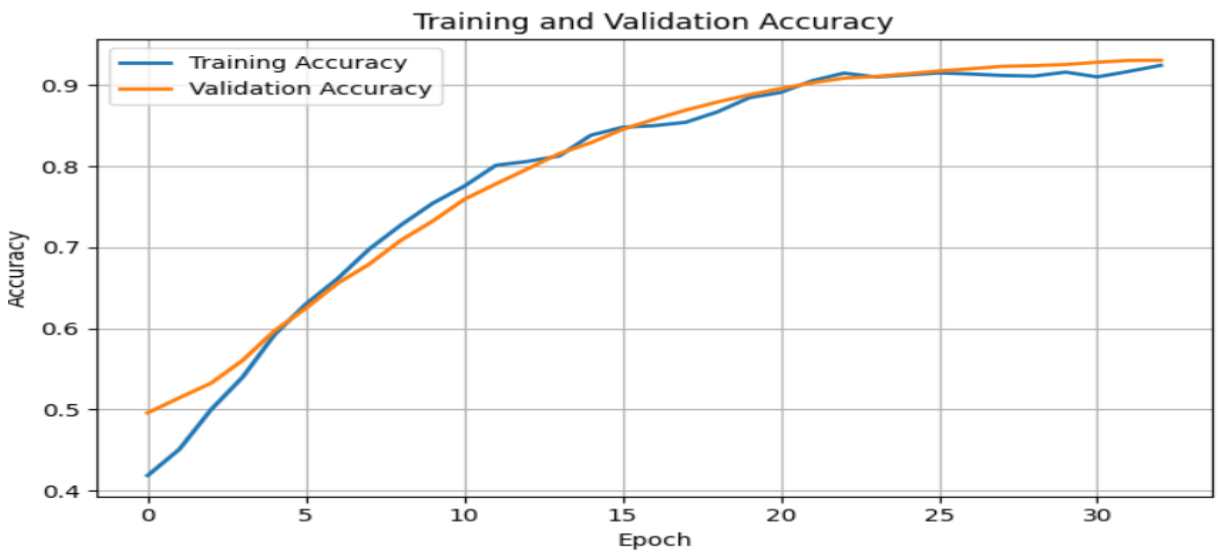

Fig. 4.1: Accuracy plot of MaxViT

From the accuracy plot, both training and validation accuracy show a consistent upward trend, indicating effective learning. The training accuracy increases from approximately 42% to over 91%, while the validation accuracy improves from around 50% to approximately 93%. The close alignment between the two curves, especially in later epochs, suggests that the model generalizes well and does not suffer from significant overfitting.

In the loss plot, both training and validation loss decrease steadily over epochs, which confirms stable convergence of the model. The training loss drops sharply from above 1.0 to nearly 0.12, while the validation loss decreases to around 0.24. Although the validation loss remains slightly higher than the training loss, the gap is minimal and consistent, indicating that the model maintains good generalization on unseen data without memorizing the training set.

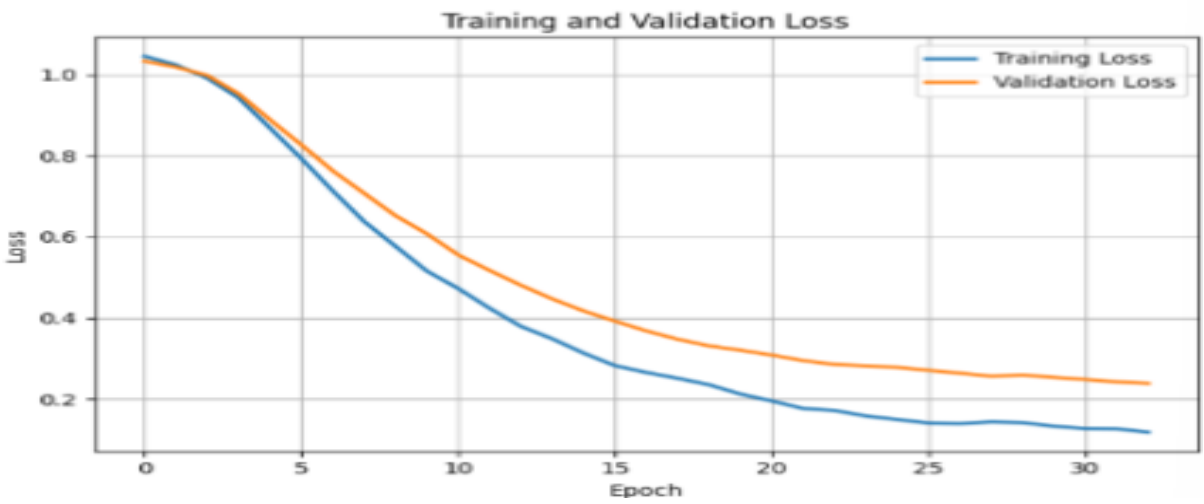

Fig. 4.2: Loss plot of MaxViT

Overall, the combined trends of increasing accuracy and decreasing loss demonstrate that the MaxViT model has successfully learned meaningful feature representations for classifying gangrene, ulcer, and normal images. The absence of large divergence between training and validation curves further indicates that the applied techniques—such as data augmentation, class balancing, and transfer learning—have effectively contributed to reducing overfitting and improving model robustness.

4.2.2 Classification Report Analysis

```
📊 Classification Report:

                      precision    recall  f1-score   support

             Gangrene     0.9184    0.9375    0.9278        48
not_ulcer_or_gangrene     1.0000    1.0000    1.0000        34
                ulcer     0.9819    0.9760    0.9790       167

             accuracy                         0.9719       249
            macro avg     0.9668    0.9712    0.9689       249
         weighted avg     0.9721    0.9719    0.9720       249

Overall Accuracy: 97.19%
```

Fig. 4.3 Classification Report of the MaxViT model

The classification report shows high precision, recall, and F1-score values for all three classes. The gangrene class achieved strong recall, indicating effective detection of severe cases. The ulcer class showed balanced precision and recall, reflecting stable performance in identifying wound conditions. The normal class achieved high precision, reducing false disease predictions.

Overall, the macro and weighted averages indicate a well-balanced and reliable classification system suitable for medical applications.

4.2.3 Confusion Matrix Analysis

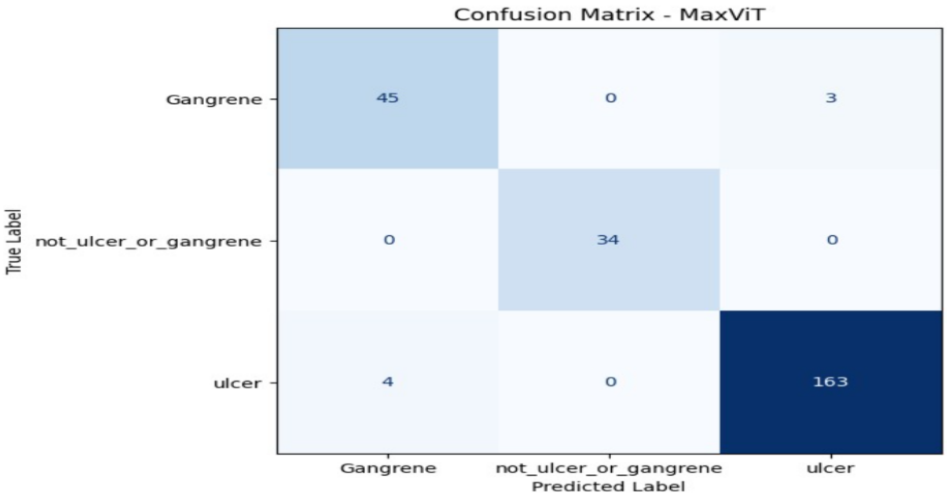

Fig. 4.4. Confusion Matrix for MaxViT Transformer

The confusion matrix analysis provides a detailed breakdown of correct and incorrect predictions across all three classes: gangrene, ulcer, and normal. The results show high diagonal dominance, indicating strong correct classification rates. Minor misclassifications occurred between ulcer and gangrene classes due to visual similarity in early-stage infections. However, overall classification performance remains highly stable and balanced across all categories.

4.2.4 ROC Curve Analysis

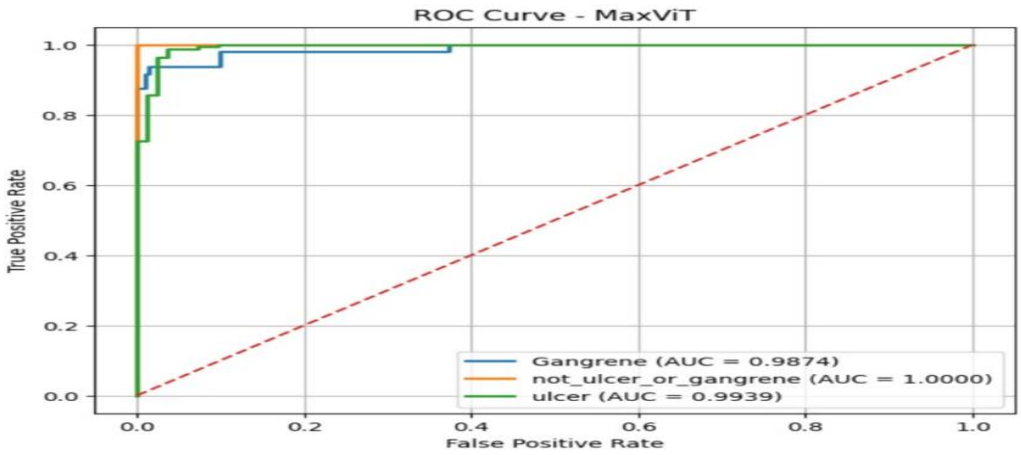

Fig. 4.5 ROC curve of the MAxViT

The ROC curve analysis demonstrates strong separability between the three classes. The model achieved high Area Under Curve (AUC) values, indicating excellent classification confidence and low false positive rates.

The curves confirm that MaxViT maintains strong performance across all threshold values, proving its robustness in multi-class medical image classification tasks.

4.2.5 Gangrene Classification using MaxViT

Fig. 4.6: Classification of gangrene using the proposed MaxViT Transformer model.

The trained MaxViT model was evaluated on the test dataset for the gangrene class. The model successfully identified gangrene images by learning distinctive visual features such as tissue discoloration, necrotic regions, and texture irregularities. The hierarchical attention mechanism of MaxViT effectively captured both local wound patterns and global structural changes present in the infected regions.

The results indicate that the model shows strong discrimination capability for gangrene images, with minimal confusion against ulcer and normal classes. This demonstrates that the transformer-based architecture is highly effective in capturing complex visual patterns associated with severe tissue damage.

4.2.6 Ulcer Classification using MaxViT

Fig. 4.7: Correct classification of ulcer using the proposed MaxViT Transformer model.

For the ulcer class, the MaxViT model demonstrated reliable performance in identifying open wound regions, redness, and surface-level skin damage. The model effectively distinguished ulcers from gangrene by focusing on differences in depth, texture, and severity of tissue damage. The classification results confirm that MaxViT can successfully learn subtle variations between ulcer and other visually similar conditions, making it suitable for early-stage wound detection systems.

4.2.7 Not Ulcer or Gangrene (Normal) Classification using MaxViT

Fig.4.8: Classification of normal using the proposed MaxViT Transformer model.

The normal class (not ulcer or gangrene) was accurately classified by the model, showing its ability to recognize healthy skin patterns. The model learned to identify the absence of infection, necrosis, or wound-like structures. This category plays a crucial role in reducing false positives, and the model performed consistently well in separating normal cases from diseased conditions, ensuring balanced multi-class classification performance.

4.2.8 Gangrene Classification with Grad-CAM++ Explainability

Fig. 4.9: Original image and corresponding Grad-CAM++ heatmap for gangrene classification.

When Grad-CAM++ was applied to gangrene predictions, heatmaps clearly highlighted infected and necrotic regions of the foot images. The highlighted areas corresponded closely to clinically relevant features such as dark tissue regions and affected skin boundaries.

This explainability confirms that the model is not making random predictions but is focusing on medically meaningful regions, thereby improving trust and interpretability of the system.

4.2.9 Ulcer Classification with Grad-CAM++ Explainability

Fig. 4.10: Original image and corresponding Grad-CAM++ heatmap for ulcer classification

Grad-CAM++ visualizations for ulcer images highlighted wound edges, inflamed regions, and open sore areas. These activation maps show that the model focuses on wound boundaries and texture variations to make predictions. The visualization improves transparency by confirming that the model bases its decisions on correct pathological regions, enhancing reliability in medical applications.

4.2.10 Normal Classification with Grad-CAM++ Explainability

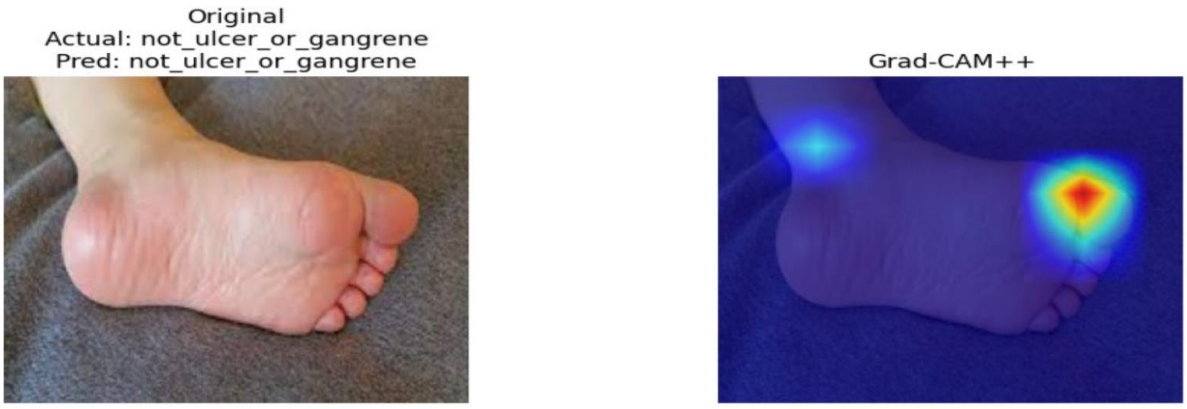

Fig. 4.11: Original image and corresponding Grad-CAM++ heatmap for not ulcer or gangrene classification.

For normal images, Grad-CAM++ showed low activation across the image, indicating that no abnormal regions were detected. The model primarily focused on general skin texture without highlighting any lesion-like regions. This behavior confirms that the model correctly identifies healthy skin conditions and avoids false detection of disease regions.

4.2.11 User Prediction – Gangrene (External Image)

Fig. 4.12: User prediction with MaxViT model(gangrene)

The model was tested on external user-provided images not included in the dataset. For gangrene cases, the model successfully generalized and correctly classified unseen images.

This demonstrates strong generalization capability and robustness of the MaxViT architecture in real-world scenarios where image conditions may vary in lighting, angle, and quality.

4.2.12 User Prediction – Ulcer (External Image)

Fig. 4.13:User prediction with MaxViT model (ulcer)

In external ulcer images, the model accurately predicted ulcer conditions even under variations such as different backgrounds and image quality. The model maintained stable performance, confirming its adaptability to real-world medical image inputs.

This indicates that the training process was effective in preventing overfitting and improving generalization.

4.2.13 User Prediction – Normal with Grad-CAM++

Fig. 4.14: User prediction with Grad-CAM++ visualization (not ulcer or gangrene)

For external normal images, the model correctly identified them as non-diseased cases. Grad-CAM++ showed minimal activation, confirming that no abnormal regions influenced the prediction. This strengthens the reliability of the system in avoiding false alarms in healthy cases.

Table 4.1 Model Comparison for Medical Image Classification

Model	Architecture Type	Key Strength	Limitation	Accuracy(%)
CNN+SVM	Hybrid (DL + ML)	Better classification using extracted CNN features	Cannot capture global context, limited interpretability	79.81
Swin Transformer	Transformer (Hierarchical Vision)	Captures global dependencies with shifted windows	High computational cost	94.12
DeiT Transformer	Vision Transformer (Data-efficient)	Efficient training with smaller datasets	Slightly less contextual learning than Swin	95.22
MaxViT	Hybrid (CNN + Multi-Axis Transformer)	Captures both local and global features effectively	Computationally intensive	97.19

The comparison clearly shows a progressive improvement in performance from traditional CNN-based models to advanced transformer-based architectures. While CNN+SVM model perform reasonably well, they are limited in capturing global spatial relationships. Transformer-based models such as Swin and DeiT significantly improve accuracy by modeling long-range dependencies. The proposed MaxViT model achieves the highest accuracy of 97.19%, demonstrating superior performance due to its ability to combine local feature extraction with global attention mechanisms.

4.3 Observations from the work

- The dataset showed real-world variations in lighting, texture, and background noise, making it suitable for robust medical image classification.
- Preprocessing (resizing to 224×224, normalization, and stratified splitting) improved training stability and ensured balanced class distribution.
- The MaxViT model effectively learned hierarchical features, starting from basic textures to complex medical patterns related to infection severity.
- AdamW optimizer with warm-up strategy and class weighting helped achieve stable convergence and balanced learning across all classes.
- The model achieved a high overall accuracy of 97.19%, demonstrating strong classification performance for gangrene, ulcer, and normal categories.
- Most misclassifications occurred between gangrene and ulcer due to their visual similarity in early-stage conditions.
- Grad-CAM++ explainability confirmed that the model focuses on medically relevant regions such as wounds, necrosis, and infected tissue areas.
- The system showed strong generalization ability on unseen test and external images, indicating good robustness.
- Transformer-based MaxViT architecture significantly improved feature representation compared to traditional CNN-based approaches.
- The model demonstrates strong potential for real-world clinical decision support,with explainability improving trust and interpretability.

CHAPTER-5

Conclusion and Future study

5.1 Conclusion

The proposed study successfully developed an explainable deep learning-based medical image classification system for identifying gangrene, ulcer, and normal (non-ulcer and non-gangrene) skin conditions. The system was built using the MaxViT architecture, which effectively combines convolutional operations with transformer-based attention mechanisms to capture both local texture details and global contextual information from medical images.

The model was trained and evaluated on a dataset of 1653 images, with appropriate preprocessing techniques such as resizing, normalization, stratified splitting, and data augmentation. The training process demonstrated stable convergence using the AdamW optimizer with warm-up and fine-tuning strategy, leading to improved learning efficiency and generalization capability.

The system achieved a final classification accuracy of 97.19%**,** indicating strong performance in distinguishing between visually similar medical conditions. The confusion matrix and classification report further confirmed balanced performance across all classes with high precision, recall, and F1-score values.

To enhance interpretability and trust in predictions, Grad-CAM++ was integrated into the system. The generated heatmaps clearly highlighted medically relevant regions such as infected tissue, wound boundaries, and necrotic areas, ensuring that the model's predictions are explainable and clinically meaningful.

Overall, the proposed system demonstrates that combining advanced transformer-based architectures with explainable AI techniques results in a reliable, accurate, and interpretable solution for automated medical image classification. This makes the system suitable for assisting healthcare professionals in early diagnosis and decision-making.

5.2 Future Study

Although the proposed system shows strong performance, there are several directions in which the work can be further improved and extended:

1. **Expansion of Dataset:**

   The model can be trained on a larger and more diverse dataset including images from different hospitals, lighting conditions, and patient demographics to improve generalization.

2. **Inclusion of More Disease Classes:**

   Future work can extend the classification to include additional skin conditions such as diabetic foot stages, infections, and other wound-related diseases.

3. **Real-Time Clinical Deployment:**

   The system can be integrated into a real-time web or mobile application to assist doctors in hospitals and remote healthcare environments.

4. **Multi-Modal Data Integration:**

   Combining medical images with patient data such as age, medical history, and symptoms can further improve prediction accuracy.

5. **Lightweight Model Optimization:**

   The MaxViT model can be optimized or compressed for deployment on edge devices such as mobile phones or portable diagnostic tools.

6. **Advanced Explainability Techniques:**

   Future improvements can include combining Grad-CAM++ with other XAI methods such as LIME or SHAP for more comprehensive interpretability.

7. **Clinical Validation:**

   The system should be validated with real clinical trials to ensure its reliability and acceptance in healthcare environments.

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APPENDIX

```python
train_tf = T.Compose([
    T.RandomResizedCrop(IMG_SIZE, scale=(0.8, 1.0)),
    T.RandomHorizontalFlip(),
    T.RandomVerticalFlip(),
    T.ColorJitter(0.2,0.2,0.2,0.1),
    T.RandomRotation(10),
    T.ToTensor(),
    T.Normalize([0.485,0.456,0.406],[0.229,0.224,0.225]),
])

val_tf = T.Compose([
    T.Resize((IMG_SIZE, IMG_SIZE)),
    T.ToTensor(),
    T.Normalize([0.485,0.456,0.406],[0.229,0.224,0.225]),
])
```

Fig. 4.15: Data Preprocessing and Augmentation

```python
model = timm.create_model(MODEL_NAME, pretrained=True, num_classes=num_classes)
model.to(device)
```

Fig. 4.16: Model Definition(MaxViT)

```python
for epoch in range(WARMUP_EPOCHS + FINETUNE_EPOCHS):

    # UNFREEZE after warmup
    if epoch == WARMUP_EPOCHS:
        for p in model.parameters():
            p.requires_grad = True
        optimizer = get_optimizer()

    model.train()
    train_loss, correct, total = 0,0,0

    loop = tqdm(train_loader, desc=f"Epoch {epoch+1}")

    for images, labels in loop:
        images, labels = images.to(device), labels.to(device)

        optimizer.zero_grad()
```

Fig. 4.17: Training Loop

```python
from sklearn.metrics import classification_report, accuracy_score

# Assuming you already have:
# all_preds, all_labels, classes

# 🔥 Classification Report (4 decimal places)
report = classification_report(
    all_labels,
    all_preds,
    target_names=classes,
    digits=4
)

# 🔥 Accuracy
acc = accuracy_score(all_labels, all_preds)

print("📊 Classification Report:\n")
print(report)

print(f"Overall Accuracy: {acc*100:.2f}%")
```

Fig. 4.18: Evaluation Metrics

```python
# Compute confusion matrix
cm = confusion_matrix(all_labels, all_preds)

# 🔥 Plot (clean style)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=classes)

fig, ax = plt.subplots(figsize=(6,6))
disp.plot(cmap="Blues", ax=ax, colorbar=False)

plt.title("Confusion Matrix - MaxViT")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.grid(False)
plt.show()
```

Fig. 4.19: Confusion Metric

```python
# ========================
# 🔥 MAKE HEATMAP SHARP (KEY PART)
# ========================

# Normalize
cam = (cam - cam.min()) / (cam.max() + 1e-8)

# 🔥 STRONG FOCUS BOOST
cam = np.power(cam, 4)   # sharper

# 🔥 REMOVE WEAK AREAS
cam = np.where(cam > 0.3, cam, 0)

# Resize
cam = cv2.resize(cam, (224,224))

# Heatmap
heatmap = cv2.applyColorMap(np.uint8(255 * cam), cv2.COLORMAP_JET)
heatmap = cv2.cvtColor(heatmap, cv2.COLOR_BGR2RGB)

# Original
img_np = np.array(img.resize((224,224)))

# 🔥 STRONG OVERLAY (VISIBLE FOCUS)
overlay = cv2.addWeighted(img_np, 0.3, heatmap, 0.7, 0)
```

Fig. 4.20: GradCAM++ Implementation

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